CERTIFICATE MATERIAL TEST REPORT

(DIN 50049.1 / EN 10204 3.1 / ISO 10474 3.1)

ISO 9001:2015

CAS METAL CO., Ltd.
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CERTIFICATE No. : CAS250605-02
CUSTOMER : NORCA INDUSTRIAL LLC
MATERIAL : ASTM/ASME A/SA182F51/60(2023ED)

ORDER NO. : PO755-005068
PROJECT NAME : P.O. N8150
ISSUED DATE : JUN. 05. 2025.

Heat No.	Spec	C	SI	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb/Cb	Al	N			
		CHEMICAL COMPOSITION (%)															
	Min.						4.5	22.0	3.0					0.14			
	Max.	0.030	1.00	2.00	0.030	0.020	6.5	23.0	3.5					0.20			
E131792	H.A. ①	0.017	0.45	1.30	0.022	0.001	5.56	22.21	3.04					0.185			
	P.A. ①																

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Non-destructive examination ⑤		
UT ⑤	N/A	-
MT ⑤	N/A	-
PT ⑤	N/A	-
Dimension & Visual Inspection	GOOD	

※ Remark

* Country of Origin : KOREA * Country of Melt & Pour : KOREA * NACE MR0175 AND MR0103

* ISO 9001 : 2015 * PED 2014 / 68 / EU. * CORROSION TESTED TO ASTM A262 PRACTICE E : No Defect

ARTICLES	Q'ty
09) B16.5 150# SO RF 8"	10EA
10) B16.5 150# SO RF 12"	40EA
11) B16.5 150# SO RF 14"	5EA
12) B16.5 150# SO RF 16"	15EA
13) B16.5 150# SO RF 18"	5EA
14) B16.5 150# SO RF 20"	5EA
15) B16.5 150# SO RF 24"	5EA
16) B16.5 150# SW RF S/40S 1"	20EA

MECHANICAL PROPERTIES	Lot No.	Location ②	Temp. ℃	TENSILE TEST ④ (Ø12.5 x 50L) Y.S. N/㎟ (MPA) MIN. 450	T.S. MIN. 655	E.L. % MIN. 25	R.A. MIN. 45	Hardness Test HBW MAX 220	CHARPY IMPACT TEST(J) Temp. : 40 °F Min. : Avg. : 1	2	3	AVG	BEND TEST Radius : Angle :
	CAS25-N24	T		491	686	42	63	162, 160	91	89	102	94	
Simulation P.W.H.T.		#1											
		#2											
		#3											

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Heat Treament ④										
	Normalizing	°C	HRS		Tempering	℃	HRS	Quenching	℃	HRS
	Solution	1040 °C	6.0 HRS	W.C	Stablization	℃	HRS	Annealing	℃	HRS

☐ **Witnessed By**

☐ **Reviewed By**

We hereby certify that the material herein has been made and tested accordance with the above specification and also with the requirement called for by the above order.

CAS METAL Co., Ltd. QUALITY CONTROL DEPART

Manager of Quality Assurance Dep't

Notes
① - H.A. = Heat Analysis, P.A. = Product Analysis
② - location : L = Longitudinal Tr = Transverse T = Tangential R = Radial Treated S.R. = Stress Relieved
③ - Surface Condition : A.F. = As Forged R.M. = Rough Machined F.M. = Finish Machined
④ - A.C. = Air Cooled, F.C. = Furnace Cooled, W.C. = Water Cooled, O.C. = Oil Cooled
⑤ - UT = Ultrasonic Test, MT = Magnetic Test, PT = Penetrant Test

ACCEPTED

BY JC DATE 1/19/26

Job: 089-25 PO: 40301 Line: 001 HT: E131792